# Solar Ca appendix

Sun · Ca · downloadable appendix

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|---|---|
| **Document** | Solar Ca appendix |
| **Subject** | Solar photospheric abundance appendix · Ca |
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Solar calcium: 6.405 ± 0.049

Asplund, Amarsi & Grevesse 2021: 6.30 ± 0.03 · this measurement +0.105 dex

The Ca I value sits about 0.1 dex above Asplund et al. 2021. They fit the light from the centre of the solar disc; we fit the light of the whole disc, as we must for other stars. Fitting the disc centre with the same lines and code gives their value, so the difference comes from the strong lines' edge-of-disc light, not from the line data. The Ca II lines agree with Asplund et al. 2021. The infrared values read about 0.1 dex higher still. On iron lines that also have laboratory values, the Cepheid-calibrated oscillator strengths sit about 0.14 dex low, which would push a calcium abundance up by about that much; the infrared products are shown as measured, with that caveat. In the red-optical only three Ca I lines can be measured (7148, 7202 and 7326 Angstrom), and they read about 0.2 to 0.3 dex below the visible lines; with so few lines those products are shown as measured but carry little weight. With the Civis et al. 2013 oscillator strengths, Ca I in the H band reads 6.18 to 6.28 on five or six lines and in the K band 6.27 to 6.37 on 19 to 22 lines, about 0.1 to 0.2 dex below the visible lines; with a 0.11 dex uncertainty on every line's oscillator strength, they are shown as measured.

Asplund, Amarsi & Grevesse (2021) give no line recipe we can repeat for this element, so this is our best HARPS product: the Reference Grade analysis with the most complete treatment, then the most lines.

VIS: 6.405 ± 0.049 · HARPS · VIS · Ca I · Asplund et al. 2021 lines · 3D NLTE (Amarsi)

## Highlighted Products

### VIS

**6.413 ± 0.048**

Ca I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · Kitt Peak (1984, molecfit) · Reference Grade

### red-optical

**6.466 ± 0.059**

Ca II · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · Kitt Peak (1984, molecfit) · Reference Grade

### NIR

**6.500 ± 0.058**

Ca I · Codex-graded lines · <3D> LTE (Gerber) · Kitt Peak (1984, molecfit) · Codex Grade

### H

**6.258 ± 0.057**

Ca I · Codex-graded lines · <3D> LTE (Gerber) · CRIRES+ H (full arm, telluric-corrected) · Codex Grade

### K

**6.365 ± 0.074**

Ca I · Codex-graded lines · <3D> LTE (Gerber) · CRIRES+ · Codex Grade

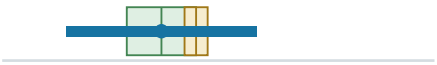
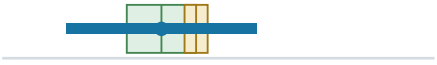
## Line grades

Every product is measured on one of three grades of spectral lines and the grades are never mixed in one number.

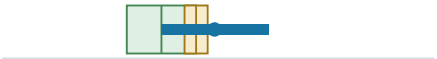
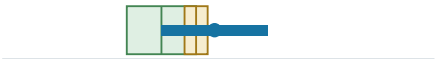
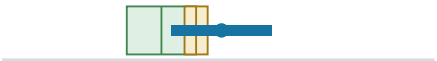
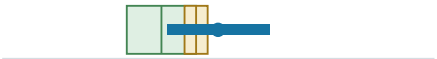
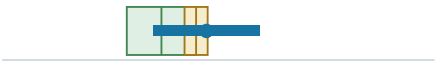
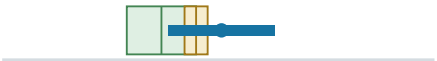
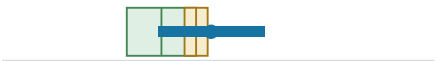
| Grade | What it means | This element |
|---|---|---|
| Reference Grade | lines from a published solar line set (Asplund et al. 2021 and similar), measured with the oscillator strengths the authors used | 102 products -- Ca I · Asplund et al. 2021 lines; Ca I · reference lines; Ca II · Asplund et al. 2021 lines |
| Codex Grade | lines chosen by the Codex whose oscillator strength carries a published uncertainty (laboratory measurement, NIST accuracy class A-C, or the paper's own stated error), between 5% and 60% deep | 110 products -- Ca I · Codex-graded lines; Ca II · Codex-graded lines |
| Deep Grade | the same oscillator-strength rule on lines deeper than 60% -- stronger lines, more sensitive to damping and the model atmosphere | none for the Sun: every line in this grade is excluded as saturated or an outlier (see Excluded lines) |

# Error-bar Forest Plot

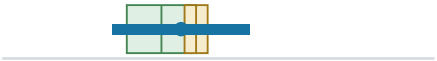
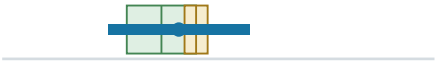
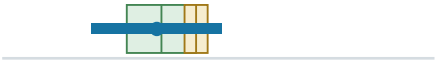
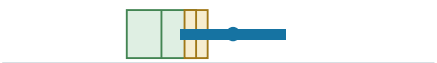
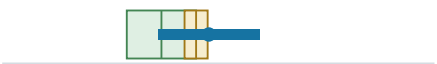
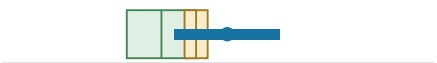
## VIS — HARPS
## A(Ca) dex; axis 6.16–6.54

| | | |
|---|---|---|
| Ca II · Asplund et al. 2021 lines · 1D LTE<br>Reference Grade · 2 lines |  | 6.293<br>±0.066 |
| Ca II · Asplund et al. 2021 lines · 3D NLTE (Amarsi)<br>Reference Grade · 2 lines |  | 6.300<br>±0.083 |
| Ca II · Asplund et al. 2021 lines · 1D NLTE (Bergemann)<br>Reference Grade · 2 lines |  | 6.290<br>±0.068 |
| Ca II · Asplund et al. 2021 lines · 1D NLTE (Gerber)<br>Reference Grade · 2 lines |  | 6.258<br>±0.066 |
| Ca II · Asplund et al. 2021 lines · 1D LTE (Gerber)<br>Reference Grade · 2 lines |  | 6.271<br>±0.063 |
| Ca II · Asplund et al. 2021 lines · <3D> LTE (Gerber)<br>Reference Grade · 2 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 6.349<br>±0.062 |
| Ca II · Asplund et al. 2021 lines · <3D> NLTE (Gerber)<br>Reference Grade · 2 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 6.332<br>±0.065 |
| Ca II · Codex-graded lines · 1D LTE<br>Codex Grade · 2 lines |  | 6.293<br>±0.066 |
| Ca II · Codex-graded lines · 3D NLTE (Amarsi)<br>Codex Grade · 2 lines |  | 6.300<br>±0.083 |
| Ca II · Codex-graded lines · 1D NLTE (Bergemann)<br>Codex Grade · 2 lines |  | 6.290<br>±0.068 |
| Ca II · Codex-graded lines · 1D NLTE (Gerber)<br>Codex Grade · 2 lines |  | 6.258<br>±0.066 |
| Ca II · Codex-graded lines · 1D LTE (Gerber)<br>Codex Grade · 2 lines |  | 6.271<br>±0.063 |
| Ca II · Codex-graded lines · <3D> LTE (Gerber)<br>Codex Grade · 2 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 6.349<br>±0.062 |
| Ca II · Codex-graded lines · <3D> NLTE (Gerber)<br>Codex Grade · 2 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 6.332<br>±0.065 |
| Ca I · Asplund et al. 2021 lines · 1D LTE<br>Reference Grade · 10 lines |  | 6.343<br>±0.046 |
| Ca I · Asplund et al. 2021 lines · 3D NLTE (Amarsi)<br>Reference Grade · 10 lines |  | 6.405<br>±0.049 |
| Ca I · Asplund et al. 2021 lines · 1D NLTE (Bergemann)<br>Reference Grade · 10 lines |  | 6.322<br>±0.045 |

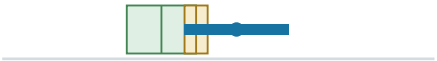
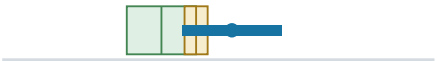
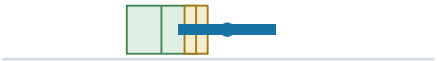
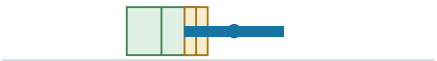
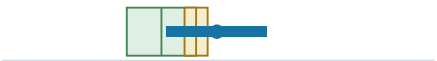
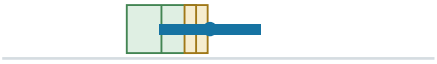
## VIS — HARPS
## A(Ca) dex; axis 6.16–6.54

| | | |
|---|---|---|
| Ca I · Asplund et al. 2021 lines · 1D NLTE (Gerber)<br>Reference Grade · 10 lines |  | 6.346<br>±0.046 |
| Ca I · Asplund et al. 2021 lines · 1D LTE (Gerber)<br>Reference Grade · 10 lines |  | 6.346<br>±0.046 |
| Ca I · Asplund et al. 2021 lines · <3D> LTE (Gerber)<br>Reference Grade · 10 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 6.438<br>±0.049 |
| Ca I · Asplund et al. 2021 lines · <3D> NLTE (Gerber)<br>Reference Grade · 10 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 6.434<br>±0.048 |
| Ca I · Codex-graded lines · 1D LTE<br>Codex Grade · 17 lines |  | 6.347<br>±0.045 |
| Ca I · Codex-graded lines · 3D NLTE (Amarsi)<br>Codex Grade · 10 lines |  | 6.405<br>±0.049 |
| Ca I · Codex-graded lines · 1D NLTE (Bergemann)<br>Codex Grade · 17 lines |  | 6.352<br>±0.043 |
| Ca I · Codex-graded lines · 1D NLTE (Gerber)<br>Codex Grade · 17 lines |  | 6.336<br>±0.044 |
| Ca I · Codex-graded lines · 1D LTE (Gerber)<br>Codex Grade · 17 lines |  | 6.349<br>±0.044 |
| Ca I · Codex-graded lines · <3D> LTE (Gerber)<br>Codex Grade · 17 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 6.438<br>±0.046 |
| Ca I · Codex-graded lines · <3D> NLTE (Gerber)<br>Codex Grade · 17 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 6.409<br>±0.044 |
| Ca I · reference lines · 1D LTE<br>Reference Grade · 45 lines |  | 6.339<br>±0.046 |
| Ca I · reference lines · 3D NLTE (Amarsi)<br>Reference Grade · 10 lines |  | 6.405<br>±0.049 |
| Ca I · reference lines · 1D NLTE (Bergemann)<br>Reference Grade · 45 lines |  | 6.352<br>±0.046 |
| Ca I · reference lines · 1D NLTE (Gerber)<br>Reference Grade · 45 lines |  | 6.310<br>±0.046 |
| Ca I · reference lines · 1D LTE (Gerber)<br>Reference Grade · 45 lines |  | 6.343<br>±0.046 |
| Ca I · reference lines · <3D> LTE (Gerber)<br>Reference Grade · 45 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 6.448<br>±0.046 |
| Ca I · reference lines · <3D> NLTE (Gerber)<br>Reference Grade · 45 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 6.411<br>±0.046 |

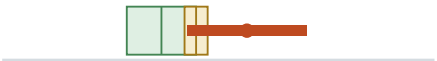
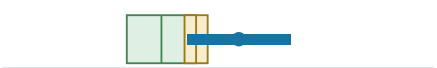
## VIS — Kitt Peak (Kurucz 2005)
## A(Ca) dex; axis 6.16–6.54

| | | |
|---|---|---|
| Ca II · Asplund et al. 2021 lines · 1D LTE<br>Reference Grade · 2 lines |  | 6.317<br>±0.059 |
| Ca II · Asplund et al. 2021 lines · 3D NLTE (Amarsi)<br>Reference Grade · 2 lines |  | 6.325<br>±0.075 |
| Ca II · Asplund et al. 2021 lines · 1D NLTE (Bergemann)<br>Reference Grade · 2 lines |  | 6.315<br>±0.061 |
| Ca II · Asplund et al. 2021 lines · 1D NLTE (Gerber)<br>Reference Grade · 2 lines |  | 6.263<br>±0.047 |
| Ca II · Asplund et al. 2021 lines · 1D LTE (Gerber)<br>Reference Grade · 2 lines |  | 6.296<br>±0.056 |
| Ca II · Asplund et al. 2021 lines · <3D> LTE (Gerber)<br>Reference Grade · 2 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 6.373<br>±0.056 |
| Ca II · Asplund et al. 2021 lines · <3D> NLTE (Gerber)<br>Reference Grade · 2 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 6.351<br>±0.058 |
| Ca II · Codex-graded lines · 1D LTE<br>Codex Grade · 2 lines |  | 6.317<br>±0.059 |
| Ca II · Codex-graded lines · 3D NLTE (Amarsi)<br>Codex Grade · 2 lines |  | 6.325<br>±0.075 |
| Ca II · Codex-graded lines · 1D NLTE (Bergemann)<br>Codex Grade · 2 lines |  | 6.315<br>±0.061 |
| Ca II · Codex-graded lines · 1D NLTE (Gerber)<br>Codex Grade · 2 lines |  | 6.263<br>±0.047 |
| Ca II · Codex-graded lines · 1D LTE (Gerber)<br>Codex Grade · 2 lines |  | 6.296<br>±0.056 |
| Ca II · Codex-graded lines · <3D> LTE (Gerber)<br>Codex Grade · 2 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 6.373<br>±0.056 |
| Ca II · Codex-graded lines · <3D> NLTE (Gerber)<br>Codex Grade · 2 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 6.351<br>±0.058 |
| Ca I · Asplund et al. 2021 lines · 1D LTE<br>Reference Grade · 10 lines |  | 6.362<br>±0.046 |
| Ca I · Asplund et al. 2021 lines · 3D NLTE (Amarsi)<br>Reference Grade · 10 lines |  | 6.412<br>±0.048 |
| Ca I · Asplund et al. 2021 lines · 1D NLTE (Bergemann)<br>Reference Grade · 10 lines |  | 6.341<br>±0.044 |
| Ca I · Asplund et al. 2021 lines · 1D NLTE (Gerber)<br>Reference Grade · 10 lines |  | 6.357<br>±0.046 |

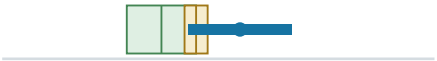
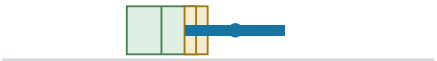
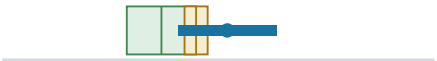
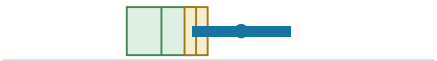
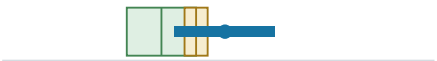
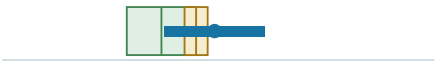
## VIS — Kitt Peak (Kurucz 2005)
## A(Ca) dex; axis 6.16–6.54

| | | |
|---|---|---|
| Ca I · Asplund et al. 2021 lines · 1D LTE (Gerber)<br>Reference Grade · 10 lines |  | 6.365<br>±0.045 |
| Ca I · Asplund et al. 2021 lines · <3D> LTE (Gerber)<br>Reference Grade · 10 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 6.455<br>±0.048 |
| Ca I · Asplund et al. 2021 lines · <3D> NLTE (Gerber)<br>Reference Grade · 10 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 6.441<br>±0.048 |
| Ca I · Codex-graded lines · 1D LTE<br>Codex Grade · 17 lines |  | 6.361<br>±0.043 |
| Ca I · Codex-graded lines · 3D NLTE (Amarsi)<br>Codex Grade · 10 lines |  | 6.412<br>±0.048 |
| Ca I · Codex-graded lines · 1D NLTE (Bergemann)<br>Codex Grade · 17 lines |  | 6.357<br>±0.042 |
| Ca I · Codex-graded lines · 1D NLTE (Gerber)<br>Codex Grade · 17 lines |  | 6.341<br>±0.043 |
| Ca I · Codex-graded lines · 1D LTE (Gerber)<br>Codex Grade · 17 lines |  | 6.363<br>±0.043 |
| Ca I · Codex-graded lines · <3D> LTE (Gerber)<br>Codex Grade · 17 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 6.449<br>±0.044 |
| Ca I · Codex-graded lines · <3D> NLTE (Gerber)<br>Codex Grade · 17 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 6.436<br>±0.044 |
| Ca I · reference lines · 1D LTE<br>Reference Grade · 45 lines |  | 6.337<br>±0.044 |
| Ca I · reference lines · 3D NLTE (Amarsi)<br>Reference Grade · 10 lines |  | 6.412<br>±0.048 |
| Ca I · reference lines · 1D NLTE (Bergemann)<br>Reference Grade · 45 lines |  | 6.348<br>±0.044 |
| Ca I · reference lines · 1D NLTE (Gerber)<br>Reference Grade · 45 lines |  | 6.298<br>±0.044 |
| Ca I · reference lines · 1D LTE (Gerber)<br>Reference Grade · 45 lines |  | 6.342<br>±0.044 |
| Ca I · reference lines · <3D> LTE (Gerber)<br>Reference Grade · 45 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 6.449<br>±0.045 |
| Ca I · reference lines · <3D> NLTE (Gerber)<br>Reference Grade · 45 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 6.402<br>±0.044 |

## VIS — Kitt Peak (1984, molecfit)
## A(Ca) dex; axis 6.16–6.54

| | | |
|---|---|---|
| Ca II · Asplund et al. 2021 lines · 1D LTE<br>Reference Grade · 2 lines |  | 6.319<br>±0.054 |
| Ca II · Asplund et al. 2021 lines · 3D NLTE (Amarsi)<br>Reference Grade · 2 lines |  | 6.327<br>±0.069 |
| Ca II · Asplund et al. 2021 lines · 1D NLTE (Bergemann)<br>Reference Grade · 2 lines |  | 6.317<br>±0.056 |
| Ca II · Asplund et al. 2021 lines · 1D NLTE (Gerber)<br>Reference Grade · 2 lines |  | 6.287<br>±0.056 |
| Ca II · Asplund et al. 2021 lines · 1D LTE (Gerber)<br>Reference Grade · 2 lines |  | 6.298<br>±0.052 |
| Ca II · Asplund et al. 2021 lines · <3D> LTE (Gerber)<br>Reference Grade · 2 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 6.374<br>±0.052 |
| Ca II · Asplund et al. 2021 lines · <3D> NLTE (Gerber)<br>Reference Grade · 2 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 6.354<br>±0.053 |
| Ca II · Codex-graded lines · 1D LTE<br>Codex Grade · 2 lines |  | 6.319<br>±0.054 |
| Ca II · Codex-graded lines · 3D NLTE (Amarsi)<br>Codex Grade · 2 lines |  | 6.327<br>±0.069 |
| Ca II · Codex-graded lines · 1D NLTE (Bergemann)<br>Codex Grade · 2 lines |  | 6.317<br>±0.056 |
| Ca II · Codex-graded lines · 1D NLTE (Gerber)<br>Codex Grade · 2 lines |  | 6.287<br>±0.056 |
| Ca II · Codex-graded lines · 1D LTE (Gerber)<br>Codex Grade · 2 lines |  | 6.298<br>±0.052 |
| Ca II · Codex-graded lines · <3D> LTE (Gerber)<br>Codex Grade · 2 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 6.374<br>±0.052 |
| Ca II · Codex-graded lines · <3D> NLTE (Gerber)<br>Codex Grade · 2 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 6.354<br>±0.053 |
| Ca I · Asplund et al. 2021 lines · 1D LTE<br>Reference Grade · 10 lines |  | 6.364<br>±0.045 |
| Ca I · Asplund et al. 2021 lines · 3D NLTE (Amarsi)<br>Reference Grade · 10 lines |  | 6.413<br>±0.048 |
| Ca I · Asplund et al. 2021 lines · 1D NLTE (Bergemann)<br>Reference Grade · 10 lines |  | 6.345<br>±0.043 |
| Ca I · Asplund et al. 2021 lines · 1D NLTE (Gerber)<br>Reference Grade · 10 lines |  | 6.367<br>±0.045 |

## VIS — Kitt Peak (1984, molecfit)
## A(Ca) dex; axis 6.16–6.54

| | | |
|---|---|---|
| Ca I · Asplund et al. 2021 lines · 1D LTE (Gerber)<br>Reference Grade · 10 lines |  | 6.368<br>±0.045 |
| Ca I · Asplund et al. 2021 lines · <3D> LTE (Gerber)<br>Reference Grade · 10 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 6.459<br>±0.047 |
| Ca I · Asplund et al. 2021 lines · <3D> NLTE (Gerber)<br>Reference Grade · 10 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 6.450<br>±0.047 |
| Ca I · Codex-graded lines · 1D LTE<br>Codex Grade · 17 lines |  | 6.364<br>±0.043 |
| Ca I · Codex-graded lines · 3D NLTE (Amarsi)<br>Codex Grade · 10 lines |  | 6.413<br>±0.048 |
| Ca I · Codex-graded lines · 1D NLTE (Bergemann)<br>Codex Grade · 17 lines |  | 6.359<br>±0.042 |
| Ca I · Codex-graded lines · 1D NLTE (Gerber)<br>Codex Grade · 17 lines |  | 6.357<br>±0.042 |
| Ca I · Codex-graded lines · 1D LTE (Gerber)<br>Codex Grade · 17 lines |  | 6.369<br>±0.043 |
| Ca I · Codex-graded lines · <3D> LTE (Gerber)<br>Codex Grade · 17 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 6.460<br>±0.044 |
| Ca I · Codex-graded lines · <3D> NLTE (Gerber)<br>Codex Grade · 17 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 6.444<br>±0.044 |
| Ca I · reference lines · 1D LTE<br>Reference Grade · 45 lines |  | 6.341<br>±0.044 |
| Ca I · reference lines · 3D NLTE (Amarsi)<br>Reference Grade · 10 lines |  | 6.413<br>±0.048 |
| Ca I · reference lines · 1D NLTE (Bergemann)<br>Reference Grade · 45 lines |  | 6.355<br>±0.043 |
| Ca I · reference lines · 1D NLTE (Gerber)<br>Reference Grade · 45 lines |  | 6.310<br>±0.043 |
| Ca I · reference lines · 1D LTE (Gerber)<br>Reference Grade · 45 lines |  | 6.346<br>±0.044 |
| Ca I · reference lines · <3D> LTE (Gerber)<br>Reference Grade · 45 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 6.462<br>±0.044 |
| Ca I · reference lines · <3D> NLTE (Gerber)<br>Reference Grade · 45 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 6.420<br>±0.044 |

VIS axis: 6.16 · 6.26 · 6.35 · 6.44 · 6.54 A(Ca) dex

## red-optical — IAG FTS
## A(Ca) dex; axis 5.85–6.62

| | | |
|---|---|---|
| Ca II · Asplund et al. 2021 lines · 1D LTE<br>Reference Grade · 2 lines |  | 6.458<br>±0.067 |
| Ca II · Asplund et al. 2021 lines · 3D NLTE (Amarsi)<br>Reference Grade · 2 lines |  | 6.472<br>±0.075 |
| Ca II · Asplund et al. 2021 lines · 1D NLTE (Bergemann)<br>Reference Grade · 2 lines |  | 6.422<br>±0.054 |
| Ca II · Asplund et al. 2021 lines · 1D NLTE (Gerber)<br>Reference Grade · 2 lines |  | 6.320<br>±0.034 |
| Ca II · Asplund et al. 2021 lines · 1D LTE (Gerber)<br>Reference Grade · 2 lines |  | 6.430<br>±0.063 |
| Ca II · Asplund et al. 2021 lines · <3D> LTE (Gerber)<br>Reference Grade · 2 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 6.485<br>±0.066 |
| Ca II · Asplund et al. 2021 lines · <3D> NLTE (Gerber)<br>Reference Grade · 2 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 6.362<br>±0.034 |
| Ca II · Codex-graded lines · 1D LTE<br>Codex Grade · 3 lines |  | 6.494<br>±0.049 |
| Ca II · Codex-graded lines · 3D NLTE (Amarsi)<br>Codex Grade · 2 lines |  | 6.472<br>±0.075 |
| Ca II · Codex-graded lines · 1D NLTE (Bergemann)<br>Codex Grade · 3 lines |  | 6.379<br>±0.046 |
| Ca II · Codex-graded lines · 1D NLTE (Gerber)<br>Codex Grade · 3 lines |  | 6.327<br>±0.034 |
| Ca II · Codex-graded lines · 1D LTE (Gerber)<br>Codex Grade · 3 lines |  | 6.464<br>±0.047 |
| Ca II · Codex-graded lines · <3D> LTE (Gerber)<br>Codex Grade · 3 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 6.523<br>±0.048 |
| Ca II · Codex-graded lines · <3D> NLTE (Gerber)<br>Codex Grade · 3 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 6.368<br>±0.034 |
| Ca I · Codex-graded lines · 1D LTE<br>Codex Grade · 3 lines |  | 6.170<br>±0.088 |
| Ca I · Codex-graded lines · 1D NLTE (Bergemann)<br>Codex Grade · 3 lines |  | 6.047<br>±0.120 |
| Ca I · Codex-graded lines · 1D NLTE (Gerber)<br>Codex Grade · 3 lines |  | 6.153<br>±0.085 |
| Ca I · Codex-graded lines · 1D LTE (Gerber)<br>Codex Grade · 3 lines |  | 6.173<br>±0.087 |

## red-optical — IAG FTS
## A(Ca) dex; axis 5.85–6.62

| | | |
|---|---|---|
| Ca I · Codex-graded lines · <3D> LTE (Gerber)<br>Codex Grade · 3 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 6.258<br>±0.090 |
| Ca I · Codex-graded lines · <3D> NLTE (Gerber)<br>Codex Grade · 3 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 6.229<br>±0.084 |
| Ca I · reference lines · 1D LTE<br>Reference Grade · 3 lines |  | 6.170<br>±0.088 |
| Ca I · reference lines · 1D NLTE (Bergemann)<br>Reference Grade · 3 lines |  | 6.047<br>±0.120 |
| Ca I · reference lines · 1D NLTE (Gerber)<br>Reference Grade · 3 lines |  | 6.153<br>±0.085 |
| Ca I · reference lines · 1D LTE (Gerber)<br>Reference Grade · 3 lines |  | 6.173<br>±0.087 |
| Ca I · reference lines · <3D> LTE (Gerber)<br>Reference Grade · 3 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 6.258<br>±0.090 |
| Ca I · reference lines · <3D> NLTE (Gerber)<br>Reference Grade · 3 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 6.229<br>±0.084 |

## red-optical — Kitt Peak (Kurucz 2005)
## A(Ca) dex; axis 5.85–6.62

| | | |
|---|---|---|
| Ca II · Asplund et al. 2021 lines · 1D LTE<br>Reference Grade · 2 lines |  | 6.477<br>±0.060 |
| Ca II · Asplund et al. 2021 lines · 3D NLTE (Amarsi)<br>Reference Grade · 2 lines |  | 6.491<br>±0.068 |
| Ca II · Asplund et al. 2021 lines · 1D NLTE (Bergemann)<br>Reference Grade · 2 lines |  | 6.441<br>±0.048 |
| Ca II · Asplund et al. 2021 lines · 1D NLTE (Gerber)<br>Reference Grade · 2 lines |  | 6.335<br>±0.037 |
| Ca II · Asplund et al. 2021 lines · 1D LTE (Gerber)<br>Reference Grade · 2 lines |  | 6.448<br>±0.056 |
| Ca II · Asplund et al. 2021 lines · <3D> LTE (Gerber)<br>Reference Grade · 2 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 6.502<br>±0.059 |
| Ca II · Asplund et al. 2021 lines · <3D> NLTE (Gerber)<br>Reference Grade · 2 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 6.377<br>±0.037 |
| Ca II · Codex-graded lines · 1D LTE<br>Codex Grade · 3 lines |  | 6.514<br>±0.046 |

## red-optical — Kitt Peak (Kurucz 2005)
## A(Ca) dex; axis 5.85–6.62

| | | |
|---|---|---|
| Ca II · Codex-graded lines · 3D NLTE (Amarsi)<br>Codex Grade · 2 lines |  | 6.491<br>±0.068 |
| Ca II · Codex-graded lines · 1D NLTE (Bergemann)<br>Codex Grade · 3 lines |  | 6.406<br>±0.043 |
| Ca II · Codex-graded lines · 1D NLTE (Gerber)<br>Codex Grade · 3 lines |  | 6.350<br>±0.036 |
| Ca II · Codex-graded lines · 1D LTE (Gerber)<br>Codex Grade · 3 lines |  | 6.485<br>±0.044 |
| Ca II · Codex-graded lines · <3D> LTE (Gerber)<br>Codex Grade · 3 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 6.544<br>±0.046 |
| Ca II · Codex-graded lines · <3D> NLTE (Gerber)<br>Codex Grade · 3 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 6.393<br>±0.035 |
| Ca I · Codex-graded lines · 1D LTE<br>Codex Grade · 3 lines |  | 6.231<br>±0.080 |
| Ca I · Codex-graded lines · 1D NLTE (Bergemann)<br>Codex Grade · 3 lines |  | 6.108<br>±0.112 |
| Ca I · Codex-graded lines · 1D NLTE (Gerber)<br>Codex Grade · 3 lines |  | 6.228<br>±0.073 |
| Ca I · Codex-graded lines · 1D LTE (Gerber)<br>Codex Grade · 3 lines |  | 6.235<br>±0.079 |
| Ca I · Codex-graded lines · <3D> LTE (Gerber)<br>Codex Grade · 3 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 6.322<br>±0.082 |
| Ca I · Codex-graded lines · <3D> NLTE (Gerber)<br>Codex Grade · 3 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 6.295<br>±0.074 |
| Ca I · reference lines · 1D LTE<br>Reference Grade · 3 lines |  | 6.231<br>±0.080 |
| Ca I · reference lines · 1D NLTE (Bergemann)<br>Reference Grade · 3 lines |  | 6.108<br>±0.112 |
| Ca I · reference lines · 1D NLTE (Gerber)<br>Reference Grade · 3 lines |  | 6.228<br>±0.073 |
| Ca I · reference lines · 1D LTE (Gerber)<br>Reference Grade · 3 lines |  | 6.235<br>±0.079 |
| Ca I · reference lines · <3D> LTE (Gerber)<br>Reference Grade · 3 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 6.322<br>±0.082 |
| Ca I · reference lines · <3D> NLTE (Gerber)<br>Reference Grade · 3 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 6.295<br>±0.074 |

## red-optical — Kitt Peak (1984, molecfit)
## A(Ca) dex; axis 5.85–6.62

| | |
|---|---|
| Ca II · Asplund et al. 2021 lines · 1D LTE<br>Reference Grade · 2 lines | 6.453<br>±0.052 |
| Ca II · Asplund et al. 2021 lines · 3D NLTE (Amarsi)<br>Reference Grade · 2 lines | 6.466<br>±0.059 |
| Ca II · Asplund et al. 2021 lines · 1D NLTE (Bergemann)<br>Reference Grade · 2 lines | 6.417<br>±0.041 |
| Ca II · Asplund et al. 2021 lines · 1D NLTE (Gerber)<br>Reference Grade · 2 lines | 6.318<br>±0.041 |
| Ca II · Asplund et al. 2021 lines · 1D LTE (Gerber)<br>Reference Grade · 2 lines | 6.425<br>±0.049 |
| Ca II · Asplund et al. 2021 lines · <3D> LTE (Gerber)<br>Reference Grade · 2 lines ·<br>EXPERIMENTAL-NOT-ADOPTED | 6.479<br>±0.051 |
| Ca II · Asplund et al. 2021 lines · <3D> NLTE (Gerber)<br>Reference Grade · 2 lines ·<br>EXPERIMENTAL-NOT-ADOPTED | 6.351<br>±0.048 |
| Ca II · Codex-graded lines · 1D LTE<br>Codex Grade · 3 lines | 6.485<br>±0.042 |
| Ca II · Codex-graded lines · 3D NLTE (Amarsi)<br>Codex Grade · 2 lines | 6.466<br>±0.059 |
| Ca II · Codex-graded lines · 1D NLTE (Bergemann)<br>Codex Grade · 3 lines | 6.392<br>±0.041 |
| Ca II · Codex-graded lines · 1D NLTE (Gerber)<br>Codex Grade · 3 lines | 6.321<br>±0.036 |
| Ca II · Codex-graded lines · 1D LTE (Gerber)<br>Codex Grade · 3 lines | 6.455<br>±0.041 |
| Ca II · Codex-graded lines · <3D> LTE (Gerber)<br>Codex Grade · 3 lines ·<br>EXPERIMENTAL-NOT-ADOPTED | 6.514<br>±0.042 |
| Ca II · Codex-graded lines · <3D> NLTE (Gerber)<br>Codex Grade · 3 lines ·<br>EXPERIMENTAL-NOT-ADOPTED | 6.383<br>±0.040 |
| Ca I · Codex-graded lines · 1D LTE<br>Codex Grade · 3 lines | 6.123<br>±0.091 |
| Ca I · Codex-graded lines · 1D NLTE (Bergemann)<br>Codex Grade · 3 lines | 6.000<br>±0.123 |
| Ca I · Codex-graded lines · 1D NLTE (Gerber)<br>Codex Grade · 3 lines | 6.121<br>±0.088 |
| Ca I · Codex-graded lines · 1D LTE (Gerber)<br>Codex Grade · 3 lines | 6.127<br>±0.090 |

## red-optical — Kitt Peak (1984, molecfit)
## A(Ca) dex; axis 5.85–6.62

| | | |
|---|---|---|
| Ca I · Codex-graded lines · <3D> LTE (Gerber)<br>Codex Grade · 3 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 6.209<br>±0.093 |
| Ca I · Codex-graded lines · <3D> NLTE (Gerber)<br>Codex Grade · 3 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 6.176<br>±0.085 |
| Ca I · reference lines · 1D LTE<br>Reference Grade · 3 lines |  | 6.123<br>±0.091 |
| Ca I · reference lines · 1D NLTE (Bergemann)<br>Reference Grade · 3 lines |  | 6.000<br>±0.123 |
| Ca I · reference lines · 1D NLTE (Gerber)<br>Reference Grade · 3 lines |  | 6.121<br>±0.088 |
| Ca I · reference lines · 1D LTE (Gerber)<br>Reference Grade · 3 lines |  | 6.127<br>±0.090 |
| Ca I · reference lines · <3D> LTE (Gerber)<br>Reference Grade · 3 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 6.209<br>±0.093 |
| Ca I · reference lines · <3D> NLTE (Gerber)<br>Reference Grade · 3 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 6.176<br>±0.085 |

red-optical axis: 5.85 · 6.04 · 6.23 · 6.43 · 6.62 A(Ca) dex

## NIR — CRIRES+ J
## A(Ca) dex; axis 6.24–6.63

| | | |
|---|---|---|
| Ca I · Codex-graded lines · 1D LTE<br>Codex Grade · 2 lines |  | 6.508<br>±0.040 |
| Ca I · Codex-graded lines · 1D NLTE (Bergemann)<br>Codex Grade · 2 lines |  | 6.502<br>±0.041 |
| Ca I · Codex-graded lines · 1D NLTE (Gerber)<br>Codex Grade · 2 lines |  | 6.473<br>±0.042 |
| Ca I · Codex-graded lines · 1D LTE (Gerber)<br>Codex Grade · 2 lines |  | 6.511<br>±0.040 |
| Ca I · Codex-graded lines · <3D> LTE (Gerber)<br>Codex Grade · 2 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 6.562<br>±0.040 |

## NIR — Kitt Peak (1984, molecfit)
## A(Ca) dex; axis 6.24–6.63

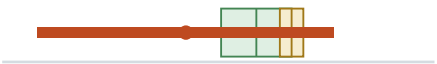
| | | |
|---|---|---|
| Ca I · Codex-graded lines · 1D LTE<br>Codex Grade · 4 lines |  | 6.439<br>±0.060 |
| Ca I · Codex-graded lines · 1D NLTE (Bergemann)<br>Codex Grade · 4 lines |  | 6.411<br>±0.070 |

## NIR — Kitt Peak (1984, molecfit)
## A(Ca) dex; axis 6.24–6.63

| | | |
|---|---|---|
| Ca I · Codex-graded lines · 1D NLTE (Gerber)<br>Codex Grade · 3 lines |  | 6.396<br>±0.072 |
| Ca I · Codex-graded lines · 1D LTE (Gerber)<br>Codex Grade · 4 lines |  | 6.441<br>±0.059 |
| Ca I · Codex-graded lines · <3D> LTE (Gerber)<br>Codex Grade · 4 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 6.500<br>±0.058 |
| Ca I · Codex-graded lines · <3D> NLTE (Gerber)<br>Codex Grade · 3 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 6.486<br>±0.070 |

NIR axis: 6.24 · 6.34 · 6.44 · 6.53 · 6.63 A(Ca) dex

## H — CRIRES+ H
## A(Ca) dex; axis 6.08–6.45

| | | |
|---|---|---|
| Ca I · Codex-graded lines · 1D LTE<br>Codex Grade · 5 lines |  | 6.236<br>±0.051 |
| Ca I · Codex-graded lines · 1D NLTE (Bergemann)<br>Codex Grade · 5 lines |  | 6.230<br>±0.050 |
| Ca I · Codex-graded lines · 1D NLTE (Gerber)<br>Codex Grade · 5 lines |  | 6.208<br>±0.051 |
| Ca I · Codex-graded lines · 1D LTE (Gerber)<br>Codex Grade · 5 lines |  | 6.250<br>±0.050 |
| Ca I · Codex-graded lines · <3D> LTE (Gerber)<br>Codex Grade · 6 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 6.283<br>±0.139 |
| Ca I · Codex-graded lines · <3D> NLTE (Gerber)<br>Codex Grade · 6 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 6.240<br>±0.126 |

## H — CRIRES+ H (full arm, telluric-corrected)
## A(Ca) dex; axis 6.08–6.45

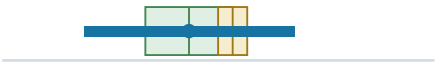
| | | |
|---|---|---|
| Ca I · Codex-graded lines · 1D LTE<br>Codex Grade · 6 lines |  | 6.198<br>±0.069 |
| Ca I · Codex-graded lines · 1D NLTE (Bergemann)<br>Codex Grade · 6 lines |  | 6.185<br>±0.067 |
| Ca I · Codex-graded lines · 1D NLTE (Gerber)<br>Codex Grade · 6 lines |  | 6.179<br>±0.063 |
| Ca I · Codex-graded lines · 1D LTE (Gerber)<br>Codex Grade · 6 lines |  | 6.204<br>±0.067 |
| Ca I · Codex-graded lines · <3D> LTE (Gerber)<br>Codex Grade · 6 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 6.258<br>±0.057 |

### H — CRIRES+ H (full arm, telluric-corrected)
### A(Ca) dex; axis 6.08–6.45

| | | |
|---|---|---|
| Ca I · Codex-graded lines · <3D> NLTE (Gerber)<br>Codex Grade · 6 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 6.209<br>±0.062 |

H axis: 6.08 · 6.18 · 6.27 · 6.36 · 6.45 A(Ca) dex

### K — CRIRES+
### A(Ca) dex; axis 6.17–6.47

| | | |
|---|---|---|
| Ca I · Codex-graded lines · 1D LTE<br>Codex Grade · 22 lines |  | 6.293<br>±0.072 |
| Ca I · Codex-graded lines · 1D NLTE (Bergemann)<br>Codex Grade · 19 lines |  | 6.286<br>±0.076 |
| Ca I · Codex-graded lines · 1D NLTE (Gerber)<br>Codex Grade · 19 lines |  | 6.270<br>±0.068 |
| Ca I · Codex-graded lines · 1D LTE (Gerber)<br>Codex Grade · 22 lines |  | 6.300<br>±0.072 |
| Ca I · Codex-graded lines · <3D> LTE (Gerber)<br>Codex Grade · 22 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 6.365<br>±0.074 |
| Ca I · Codex-graded lines · <3D> NLTE (Gerber)<br>Codex Grade · 19 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 6.297<br>±0.072 |

K axis: 6.17 · 6.25 · 6.32 · 6.39 · 6.47 A(Ca) dex

**How to read this forest.** Each row is one independent product, preserving its band, holding, engine, indicator and line set. Engines and indicator families are never averaged. Each bar is the uncertainty as Asplund et al. 2021 quote it: line-to-line scatter, the 3D-model term and the NLTE term (at least 0.03 dex). For a molecule the NLTE term is replaced by its dependence on the other element, and the 3D term comes from the 3D-to-1D difference Amarsi et al. 2021 publish for the same band. Atomic-data and observational terms are not in the bar, as in Asplund; they stay in the full error budget attached to the row. Grades describe the supplied line-pool provenance, not engine quality; evaluated theoretical CNO oscillator strengths are not relabelled as laboratory evidence. Quarantined, failed and unconstrained fits have no abundance mark. Experimental products remain explicitly not adopted. Literature references provide context, never tuning targets.

## How this was measured

Every calcium value on this page comes from fitting synthetic spectra to the observed solar spectrum, line by line or band by band, with the calcium abundance as the free parameter. The synthesis includes the atomic lines and the molecular opacity in every window, so blended absorption is modelled rather than attributed to the line being measured. The continuum is placed locally around every line by comparing the observation with the synthesis on its own highest pixels, and ground-based spectra are telluric-corrected across their full range.

Each indicator is measured independently on each solar spectrum and reported separately; nothing is averaged across indicators or instruments. Where a 3D or NLTE correction is published for the lines, it is applied as its own product (1D-LTE, 1D-NLTE and 3D-NLTE are separate rows). Every value carries a full uncertainty budget: fit, line strengths, microturbulence, continuum, fit window, model atmosphere, telluric residual, and the spread between spectra, each measured by refitting with that one input changed. The headline at the top is the Asplund-grade product: a published line set, the most complete treatment, then the most lines.

| | |
|---|---|
| Band | VIS |
| What was measured | Ca I · Codex-graded lines; Ca II · Codex-graded lines |
| Solar spectra | HARPS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005) |
| Treatments | 1D-LTE, 1D-NLTE, 3D-NLTE, synth-mean3D-LTE-gerber-stagger, synth-mean3D-NLTE-gerber-stagger |
| Lines | 2–17 |
| Result A(Ca) | 6.300–6.413<br>(3D-NLTE) |
| Largest uncertainty terms (dex) | NLTE / 3D correction 0.03, model atmosphere 0.03 |

| | |
|---|---|
| Band | VIS |
| What was measured | Ca I · reference lines |
| Solar spectra | HARPS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005) |
| Treatments | 1D-LTE, 1D-NLTE, 3D-NLTE, synth-mean3D-LTE-gerber-stagger, synth-mean3D-NLTE-gerber-stagger |
| Lines | 10–45 |
| Result A(Ca) | 6.405–6.413<br>(3D-NLTE) |
| Largest uncertainty terms (dex) | NLTE / 3D correction 0.03, model atmosphere 0.03 |

| | |
|---|---|
| Band | VIS |
| What was measured | Ca I · Asplund et al. 2021 lines; Ca II · Asplund et al. 2021 lines |
| Solar spectra | HARPS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005) |
| Treatments | 1D-LTE, 1D-NLTE, 3D-NLTE, synth-mean3D-LTE-gerber-stagger, synth-mean3D-NLTE-gerber-stagger |
| Lines | 2–10 |
| Result A(Ca) | 6.300–6.413<br>(3D-NLTE) |
| Largest uncertainty terms (dex) | NLTE / 3D correction 0.03, model atmosphere 0.03 |

| | |
|---|---|
| Band | red-optical |
| What was measured | Ca I · Codex-graded lines; Ca II · Codex-graded lines |
| Solar spectra | IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005) |
| Treatments | 1D-LTE, 1D-NLTE, 3D-NLTE, synth-mean3D-LTE-gerber-stagger, synth-mean3D-NLTE-gerber-stagger |
| Lines | 2–3 |
| Result A(Ca) | 6.466–6.491<br>(3D-NLTE) |
| Largest uncertainty terms (dex) | line strengths (gf) 0.07, fit / line-to-line scatter 0.05 |

| | |
|---|---|
| Band | red-optical |
| What was measured | Ca I · reference lines |
| Solar spectra | IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005) |
| Treatments | 1D-LTE, 1D-NLTE, synth-mean3D-LTE-gerber-stagger, synth-mean3D-NLTE-gerber-stagger |
| Lines | 3 |
| Result A(Ca) | 6.176–6.295<br>(synth-mean3D-NLTE-gerber-stagger) |
| Largest uncertainty terms (dex) | fit / line-to-line scatter 0.06, model atmosphere 0.03 |

| | |
|---|---|
| Band | red-optical |
| What was measured | Ca II · Asplund et al. 2021 lines |
| Solar spectra | IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005) |
| Treatments | 1D-LTE, 1D-NLTE, 3D-NLTE, synth-mean3D-LTE-gerber-stagger, synth-mean3D-NLTE-gerber-stagger |
| Lines | 2 |
| Result A(Ca) | 6.466–6.491<br>(3D-NLTE) |
| Largest uncertainty terms (dex) | line strengths (gf) 0.07, fit / line-to-line scatter 0.05 |

| | |
|---|---|
| Band | NIR |
| What was measured | Ca I · Codex-graded lines |
| Solar spectra | CRIRES+ J, Kitt Peak (1984, molecfit) |
| Treatments | 1D-LTE, 1D-NLTE, synth-mean3D-LTE-gerber-stagger, synth-mean3D-NLTE-gerber-stagger |
| Lines | 2–4 |
| Result A(Ca) | 6.500–6.562<br>(synth-mean3D-LTE-gerber-stagger) |
| Largest uncertainty terms (dex) | line strengths (gf) 0.13, fit / line-to-line scatter 0.04 |

| | |
|---|---|
| Band | H |
| What was measured | Ca I · Codex-graded lines |
| Solar spectra | CRIRES+ H, CRIRES+ H (full arm, telluric-corrected) |
| Treatments | 1D-LTE, 1D-NLTE, synth-mean3D-LTE-gerber-stagger, synth-mean3D-NLTE-gerber-stagger |
| Lines | 5–6 |
| Result A(Ca) | 6.258–6.283<br>(synth-mean3D-LTE-gerber-stagger) |
| Largest uncertainty terms (dex) | line strengths (gf) 0.11, fit / line-to-line scatter 0.04 |

| | |
|---|---|
| Band | K |
| What was measured | Ca I · Codex-graded lines |
| Solar spectra | CRIRES+ |
| Treatments | 1D-LTE, 1D-NLTE, synth-mean3D-LTE-gerber-stagger, synth-mean3D-NLTE-gerber-stagger |
| Lines | 19–22 |
| Result A(Ca) | 6.365<br>(synth-mean3D-LTE-gerber-stagger) |
| Largest uncertainty terms (dex) | line strengths (gf) 0.11, fit / line-to-line scatter 0.06 |

## Excluded lines

These lines are left out of the Sun's measurement. They stay in the line lists: a cooler or more evolved star may measure them well. Blended lines are not excluded -- the spectrum synthesis models the blend. 5 lines.

| Species | Wavelength (Å, air) | Why it is excluded |
|---|---|---|
| Ca I | 4307.744 | molecular_blend |
| Ca I | 4434.957 | atomic_blend |
| Ca I | 4526.928 | atomic_blend |

| Species | Wavelength (Å, air) | Why it is excluded |
|---|---|---|
| Ca I | 5041.618 | atomic_blend |
| Ca I | 5270.270 | atomic_blend |

## Solar calcium context

Calcium is measured from neutral Ca I and ionized Ca II lines in the visible and red-optical. The reference lines are the ten Ca I and five Ca II lines Asplund et al. 2021 used, with the oscillator strengths they adopted; a wider set of lines with laboratory oscillator strengths and the Codex's own line selection are shown as separate products. In the infrared (Y and J bands, Kitt Peak and CRIRES+) no laboratory oscillator strength exists, so the lines use values Elgueta et al. 2024 calibrated on Cepheid stars, each with its own published uncertainty. In the H and K bands (CRIRES+) the only values not tuned to the Sun are ones Civis et al. 2013 calculated from atomic theory. We tested them: their magnesium values match our laboratory-graded magnesium lines to about 0.04 dex, and on the two calcium lines they share with the Cepheid calibration they differ by 0.11 dex, which we carry as each line's uncertainty. The values Kurucz computed, which we used before, are about 0.16 dex too strong for the main H-band lines and read calcium low by that much. No ionized calcium line in H or K is deep enough to measure, and no usable calcium line was found in the near-ultraviolet.

Published line sets · gf uncertainty model · Methodology

## Download the evidence

Download PDF — Solar Ca appendix

Source feed — includes quarantined fit outputs, not adopted abundances

Complete product visibility audit

Shared website/PDF report

Reproducibility manifest

Complete references and product-to-source bindings

## Reproducibility

Generated from the pinned science snapshot. Source commit 1160d75f5d7cefc4023130d56363d6d18c21cdea
Feed Ca.json v1.369 · updated 2026-11-07T22:07:32Z
Generated 2026-11-07T15:10:37-07:00 (source commit timestamp)
Feed SHA-256 85ca0dfa2046775ad7457bf2e32e41bc75ef097a79edb6ca0e35b148f5328608
Generator scripts/generate_cno_publication.py v1.0.0

Pinned scientific source

## References / Data & Model Sources

Sources are separated by their role. Registered grid lineage and quarantined diagnostics do not imply an adopted abundance or an applied correction.

See the source notes for the distinction between reference lineage and the treatment actually recorded in each feed entry.

Science and provenance source notes

**Solar abundance and indicator analyses.** Solar Ca comparison literature; not an adopted Codex abundance. Sources: [1], [2].

**Solar abundance and indicator analyses.** Published line set CA_AGSS21 (the line selection). Sources: [1].

**Atomic data and gf authorities.** Ca II oscillator strengths of this pool, as the product's own uncertainty budget records them (with their lineage). Sources: [1].

**Model and grid sources.** Feed treatment: 1D-LTE synthesis (Turbospectrum); no NLTE/3D correction claimed. Sources: [3].

**Model and grid sources.** Cited solar abundance differences from the published model columns, applied additively to our LTE fits; not a new in-transfer 3D calculation. Baseline atmosphere: A_marcs_lte Sources: [1].

**Model and grid sources.** Feed treatment: ENGINE-B-NLTE; Turbospectrum on the registered Gerber deck. The product retains its own atmosphere and ion identity. Sources: [3].

**Model and grid sources.** Feed treatment: synth-1D-LTE-gerber; Turbospectrum on the registered Gerber deck. The product retains its own atmosphere and ion identity. Sources: [3].

**Model and grid sources.** Feed treatment: synth-mean3D-LTE-gerber-stagger; Turbospectrum on the registered Gerber deck with the horizontally averaged 3D (<3D>) Stagger solar model; not a full 3D calculation. Sources: [3], [4].

**Model and grid sources.** Feed treatment: synth-mean3D-NLTE-gerber-stagger; Turbospectrum on the registered Gerber deck with the horizontally averaged 3D (<3D>) Stagger solar model; not a full 3D calculation. Sources: [3], [4].

**Atomic data and gf authorities.** Ca I oscillator strengths of this pool, as the product's own uncertainty budget records them (with their lineage). Sources: [6].

**Model and grid sources.** Feed treatment: 1D NLTE, per-line correction from the Amarsi 2020 GALAH departure grid via PySME (data/nlte_grids/Ca_Amarsi2020_PySME.csv). Sources: [7].

Product-to-source index

VIS · HARPS · Ca II · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3] Ca-products-dd4060b496e3

VIS · HARPS · Ca II · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · [1], [2] Ca-products-24ee92a42efb

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H · CRIRES+ H · Ca I · Codex-graded lines · 1D LTE · [1], [2], [6], [3] Ca-products-db8c4a4577ed

H · CRIRES+ H · Ca I · Codex-graded lines · 1D NLTE (Bergemann) · [1], [2], [6], [7] Ca-products-8c3fc50ea2a8

H · CRIRES+ H · Ca I · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [6], [3] Ca-products-dc19151fc76d

H · CRIRES+ H · Ca I · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [6], [3] Ca-products-df48bfb8047c

H · CRIRES+ H · Ca I · Codex-graded lines · <3D> LTE (Gerber) · [1], [2], [6], [3], [4] Ca-products-b83f252cfd0b

H · CRIRES+ H · Ca I · Codex-graded lines · <3D> NLTE (Gerber) · [1], [2], [6], [3], [4] Ca-products-646286a13931

H · CRIRES+ H (full arm, telluric-corrected) · Ca I · Codex-graded lines · 1D LTE · [1], [2], [6], [3] Ca-products-94e084040d7a

H · CRIRES+ H (full arm, telluric-corrected) · Ca I · Codex-graded lines · 1D NLTE (Bergemann) · [1], [2], [6], [7] Ca-products-a11e63f7f4b0

H · CRIRES+ H (full arm, telluric-corrected) · Ca I · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [6], [3] Ca-products-f730f931ae70

H · CRIRES+ H (full arm, telluric-corrected) · Ca I · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [6], [3] Ca-products-e8c904a8b6bf

H · CRIRES+ H (full arm, telluric-corrected) · Ca I · Codex-graded lines · <3D> LTE (Gerber) · [1], [2], [6], [3], [4] Ca-products-af2d80436592

H · CRIRES+ H (full arm, telluric-corrected) · Ca I · Codex-graded lines · <3D> NLTE (Gerber) · [1], [2], [6], [3], [4] Ca-products-868bc40ff570

K · CRIRES+ · Ca I · Codex-graded lines · 1D LTE · [1], [2], [6], [3] Ca-products-035b072011ba

K · CRIRES+ · Ca I · Codex-graded lines · 1D NLTE (Bergemann) · [1], [2], [6], [7] Ca-products-a9ba07fc271e

K · CRIRES+ · Ca I · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [6], [3] Ca-products-0c2c577d3f3d

K · CRIRES+ · Ca I · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [6], [3] Ca-products-1a15de6b4f92

K · CRIRES+ · Ca I · Codex-graded lines · <3D> LTE (Gerber) · [1], [2], [6], [3], [4] Ca-products-1392e7a95591

K · CRIRES+ · Ca I · Codex-graded lines · <3D> NLTE (Gerber) · [1], [2], [6], [3], [4] Ca-products-37a878b7743f

NIR · CRIRES+ J · Ca I · Codex-graded lines · 1D LTE · [1], [2], [8], [3] Ca-products-5a96bdcb5ca8

NIR · CRIRES+ J · Ca I · Codex-graded lines · 1D NLTE (Bergemann) · [1], [2], [8], [7] Ca-products-9d1e63e12848

NIR · CRIRES+ J · Ca I · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [8], [3] Ca-products-2b81e1c6dff3

NIR · CRIRES+ J · Ca I · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [8], [3] Ca-products-bba1e2a3498f

NIR · CRIRES+ J · Ca I · Codex-graded lines · <3D> LTE (Gerber) · [1], [2], [8], [3], [4] Ca-products-bf620e648cd9

NIR · CRIRES+ J · Ca I · Codex-graded lines · <3D> NLTE (Gerber) · [1], [2], [8], [3], [4] Ca-products-09938451604f

NIR · Kitt Peak (1984, molecfit) · Ca I · Codex-graded lines · 1D LTE · [1], [2], [8], [3] Ca-products-91d5e1aec0ed

NIR · Kitt Peak (1984, molecfit) · Ca I · Codex-graded lines · 1D NLTE (Bergemann) · [1], [2], [8], [7] Ca-products-60c180270e91

NIR · Kitt Peak (1984, molecfit) · Ca I · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [8], [3] Ca-products-ed08e0518d30

NIR · Kitt Peak (1984, molecfit) · Ca I · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [8], [3] Ca-products-78b9cb40507e

NIR · Kitt Peak (1984, molecfit) · Ca I · Codex-graded lines · <3D> LTE (Gerber) · [1], [2], [8], [3], [4] Ca-products-ba8be72112a0

NIR · Kitt Peak (1984, molecfit) · Ca I · Codex-graded lines · <3D> NLTE (Gerber) · [1], [2], [8], [3], [4] Ca-products-6b51a2fdc773

VIS · HARPS · Ca I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [9], [3] Ca-products-0fd16bf7954c

VIS · HARPS · Ca I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · [1], [2], [9] Ca-products-1730bf153fa7

VIS · HARPS · Ca I · Asplund et al. 2021 lines · 1D NLTE (Bergemann) · [1], [2], [9] Ca-products-88794028e3a2

VIS · HARPS · Ca I · Asplund et al. 2021 lines · 1D NLTE (Gerber) · [1], [2], [9], [3] Ca-products-879a1400546d

VIS · HARPS · Ca I · Asplund et al. 2021 lines · 1D LTE (Gerber) · [1], [2], [9], [3] Ca-products-54e68ce4264d

VIS · HARPS · Ca I · Asplund et al. 2021 lines · <3D> LTE (Gerber) · [1], [2], [9], [3], [4] Ca-products-707105a245f8

VIS · HARPS · Ca I · Asplund et al. 2021 lines · <3D> NLTE (Gerber) · [1], [2], [9], [3], [4] Ca-products-7199d5d676c4

VIS · HARPS · Ca I · Codex-graded lines · 1D LTE · [1], [2], [9], [3] Ca-products-9c9e25eeaa78

VIS · HARPS · Ca I · Codex-graded lines · 3D NLTE (Amarsi) · [1], [2], [9] Ca-products-42a24832a7ec

VIS · HARPS · Ca I · Codex-graded lines · 1D NLTE (Bergemann) · [1], [2], [9] Ca-products-dfe8ec67c3a1

VIS · HARPS · Ca I · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [9], [3] Ca-products-aec980e50ce3

VIS · HARPS · Ca I · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [9], [3] Ca-products-03b41cb706e2

VIS · HARPS · Ca I · Codex-graded lines · <3D> LTE (Gerber) · [1], [2], [9], [3], [4] Ca-products-9ab995181130

VIS · HARPS · Ca I · Codex-graded lines · <3D> NLTE (Gerber) · [1], [2], [9], [3], [4] Ca-products-7845bc32873d

VIS · HARPS · Ca I · reference lines · 1D LTE · [1], [2], [9], [3] Ca-products-0bd223136c0f

VIS · HARPS · Ca I · reference lines · 3D NLTE (Amarsi) · [1], [2], [9] Ca-products-89ca56c86618

VIS · HARPS · Ca I · reference lines · 1D NLTE (Bergemann) · [1], [2], [9] Ca-products-69c22c7e5631

VIS · HARPS · Ca I · reference lines · 1D NLTE (Gerber) · [1], [2], [9], [3] Ca-products-d1504b40bf1c

VIS · HARPS · Ca I · reference lines · 1D LTE (Gerber) · [1], [2], [9], [3] Ca-products-b3b308b89591

VIS · HARPS · Ca I · reference lines · <3D> LTE (Gerber) · [1], [2], [9], [3], [4] Ca-products-ba068985e996

VIS · HARPS · Ca I · reference lines · <3D> NLTE (Gerber) · [1], [2], [9], [3], [4] Ca-products-fdd873b81f1f

VIS · Kitt Peak (Kurucz 2005) · Ca I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [9], [3] Ca-products-c7f81e52ea77

VIS · Kitt Peak (Kurucz 2005) · Ca I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · [1], [2], [9] Ca-products-78de95be737a

VIS · Kitt Peak (Kurucz 2005) · Ca I · Asplund et al. 2021 lines · 1D NLTE (Bergemann) · [1], [2], [9] Ca-products-6ccd120a151c

VIS · Kitt Peak (Kurucz 2005) · Ca I · Asplund et al. 2021 lines · 1D NLTE (Gerber) · [1], [2], [9], [3] Ca-products-3a977e428f38

VIS · Kitt Peak (Kurucz 2005) · Ca I · Asplund et al. 2021 lines · 1D LTE (Gerber) · [1], [2], [9], [3] Ca-products-329d1d51a94a

VIS · Kitt Peak (Kurucz 2005) · Ca I · Asplund et al. 2021 lines · <3D> LTE (Gerber) · [1], [2], [9], [3], [4] Ca-products-4cc273e34c25

VIS · Kitt Peak (Kurucz 2005) · Ca I · Asplund et al. 2021 lines · <3D> NLTE (Gerber) · [1], [2], [9], [3], [4] Ca-products-89504ceb38ac

VIS · Kitt Peak (Kurucz 2005) · Ca I · Codex-graded lines · 1D LTE · [1], [2], [9], [3] Ca-products-89586ce7bde2

VIS · Kitt Peak (Kurucz 2005) · Ca I · Codex-graded lines · 3D NLTE (Amarsi) · [1], [2], [9] Ca-products-3e6ff73e566a

VIS · Kitt Peak (Kurucz 2005) · Ca I · Codex-graded lines · 1D NLTE (Bergemann) · [1], [2], [9] Ca-products-0906a081ca3e

VIS · Kitt Peak (Kurucz 2005) · Ca I · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [9], [3] Ca-products-5eaf97a9a943

VIS · Kitt Peak (Kurucz 2005) · Ca I · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [9], [3] Ca-products-9c0b0a71c34a

VIS · Kitt Peak (Kurucz 2005) · Ca I · Codex-graded lines · <3D> LTE (Gerber) · [1], [2], [9], [3], [4] Ca-products-30e55118bf57

VIS · Kitt Peak (Kurucz 2005) · Ca I · Codex-graded lines · <3D> NLTE (Gerber) · [1], [2], [9], [3], [4] Ca-products-2b245c9a0fb1

VIS · Kitt Peak (Kurucz 2005) · Ca I · reference lines · 1D LTE · [1], [2], [9], [3] Ca-products-80133bcb2a24

VIS · Kitt Peak (Kurucz 2005) · Ca I · reference lines · 3D NLTE (Amarsi) · [1], [2], [9] Ca-products-02854f85bb1b

VIS · Kitt Peak (Kurucz 2005) · Ca I · reference lines · 1D NLTE (Bergemann) · [1], [2], [9] Ca-products-f625c2d73713

VIS · Kitt Peak (Kurucz 2005) · Ca I · reference lines · 1D NLTE (Gerber) · [1], [2], [9], [3] Ca-products-b7419f3d8634

VIS · Kitt Peak (Kurucz 2005) · Ca I · reference lines · 1D LTE (Gerber) · [1], [2], [9], [3] Ca-products-d3729bcb0579

VIS · Kitt Peak (Kurucz 2005) · Ca I · reference lines · <3D> LTE (Gerber) · [1], [2], [9], [3], [4] Ca-products-4965d0817cbd

VIS · Kitt Peak (Kurucz 2005) · Ca I · reference lines · <3D> NLTE (Gerber) · [1], [2], [9], [3], [4] Ca-products-d566b4acebd6

VIS · Kitt Peak (1984, molecfit) · Ca I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [9], [3] Ca-products-6581b9617b38

VIS · Kitt Peak (1984, molecfit) · Ca I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · [1], [2], [9] Ca-products-7c4323931f2e

VIS · Kitt Peak (1984, molecfit) · Ca I · Asplund et al. 2021 lines · 1D NLTE (Bergemann) · [1], [2], [9] Ca-products-276d47ee4037

VIS · Kitt Peak (1984, molecfit) · Ca I · Asplund et al. 2021 lines · 1D NLTE (Gerber) · [1], [2], [9], [3] Ca-products-0e4ef155a4b7

VIS · Kitt Peak (1984, molecfit) · Ca I · Asplund et al. 2021 lines · 1D LTE (Gerber) · [1], [2], [9], [3] Ca-products-451beddb48fc

VIS · Kitt Peak (1984, molecfit) · Ca I · Asplund et al. 2021 lines · <3D> LTE (Gerber) · [1], [2], [9], [3], [4] Ca-products-ca6a2b787d90

VIS · Kitt Peak (1984, molecfit) · Ca I · Asplund et al. 2021 lines · <3D> NLTE (Gerber) · [1], [2], [9], [3], [4] Ca-products-966966d6d683

VIS · Kitt Peak (1984, molecfit) · Ca I · Codex-graded lines · 1D LTE · [1], [2], [9], [3] Ca-products-5cba243aad5b

VIS · Kitt Peak (1984, molecfit) · Ca I · Codex-graded lines · 3D NLTE (Amarsi) · [1], [2], [9] Ca-products-6a0a8bcda2e6

VIS · Kitt Peak (1984, molecfit) · Ca I · Codex-graded lines · 1D NLTE (Bergemann) · [1], [2], [9] Ca-products-872ff4e5250d

VIS · Kitt Peak (1984, molecfit) · Ca I · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [9], [3] Ca-products-7de3ddeed508

VIS · Kitt Peak (1984, molecfit) · Ca I · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [9], [3] Ca-products-168927ebf6ef

VIS · Kitt Peak (1984, molecfit) · Ca I · Codex-graded lines · <3D> LTE (Gerber) · [1], [2], [9], [3], [4] Ca-products-57377029b850

VIS · Kitt Peak (1984, molecfit) · Ca I · Codex-graded lines · <3D> NLTE (Gerber) · [1], [2], [9], [3], [4] Ca-products-a7375edd7aa4

VIS · Kitt Peak (1984, molecfit) · Ca I · reference lines · 1D LTE · [1], [2], [9], [3] Ca-products-317e37cfc90d

VIS · Kitt Peak (1984, molecfit) · Ca I · reference lines · 3D NLTE (Amarsi) · [1], [2], [9] Ca-products-7e73a77b8fe3

VIS · Kitt Peak (1984, molecfit) · Ca I · reference lines · 1D NLTE (Bergemann) · [1], [2], [9] Ca-products-1e62f59f94d2

VIS · Kitt Peak (1984, molecfit) · Ca I · reference lines · 1D NLTE (Gerber) · [1], [2], [9], [3] Ca-products-f757ff241a89

VIS · Kitt Peak (1984, molecfit) · Ca I · reference lines · 1D LTE (Gerber) · [1], [2], [9], [3] Ca-products-075e01abd2f9

VIS · Kitt Peak (1984, molecfit) · Ca I · reference lines · <3D> LTE (Gerber) · [1], [2], [9], [3], [4] Ca-products-4560ecdacb4b

VIS · Kitt Peak (1984, molecfit) · Ca I · reference lines · <3D> NLTE (Gerber) · [1], [2], [9], [3], [4] Ca-products-da17ea6f3f64

red-optical · IAG FTS · Ca I · Codex-graded lines · 1D LTE · [1], [2], [9], [3] Ca-products-5d9595047f2c

red-optical · IAG FTS · Ca I · Codex-graded lines · 1D NLTE (Bergemann) · [1], [2], [9], [7] Ca-products-75f058b0cff5

red-optical · IAG FTS · Ca I · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [9], [3] Ca-products-5bc736a79d60

red-optical · IAG FTS · Ca I · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [9], [3] Ca-products-84df940e996c

red-optical · IAG FTS · Ca I · Codex-graded lines · <3D> LTE (Gerber) · [1], [2], [9], [3], [4] Ca-products-bfe0d443cb86

red-optical · IAG FTS · Ca I · Codex-graded lines · <3D> NLTE (Gerber) · [1], [2], [9], [3], [4] Ca-products-0e70b718f154

red-optical · IAG FTS · Ca I · reference lines · 1D LTE · [1], [2], [9], [3] Ca-products-5afe6e142880

red-optical · IAG FTS · Ca I · reference lines · 1D NLTE (Bergemann) · [1], [2], [9], [7] Ca-products-dc14e6d0a86d

red-optical · IAG FTS · Ca I · reference lines · 1D NLTE (Gerber) · [1], [2], [9], [3] Ca-products-0033b75c2433

red-optical · IAG FTS · Ca I · reference lines · 1D LTE (Gerber) · [1], [2], [9], [3] Ca-products-0c9748102733

red-optical · IAG FTS · Ca I · reference lines · <3D> LTE (Gerber) · [1], [2], [9], [3], [4] Ca-products-3f3602a94c36

red-optical · IAG FTS · Ca I · reference lines · <3D> NLTE (Gerber) · [1], [2], [9], [3], [4] Ca-products-523ca8472cdf

red-optical · Kitt Peak (Kurucz 2005) · Ca I · Codex-graded lines · 1D LTE · [1], [2], [9], [3] Ca-products-3475fb9b54e3

red-optical · Kitt Peak (Kurucz 2005) · Ca I · Codex-graded lines · 1D NLTE (Bergemann) · [1], [2], [9], [7] Ca-products-95f84e301ace

red-optical · Kitt Peak (Kurucz 2005) · Ca I · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [9], [3] Ca-products-cf6b24cd86c6

red-optical · Kitt Peak (Kurucz 2005) · Ca I · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [9], [3] Ca-products-f697331d17eb

red-optical · Kitt Peak (Kurucz 2005) · Ca I · Codex-graded lines · <3D> LTE (Gerber) · [1], [2], [9], [3], [4] Ca-products-6cde383fa798

red-optical · Kitt Peak (Kurucz 2005) · Ca I · Codex-graded lines · <3D> NLTE (Gerber) · [1], [2], [9], [3], [4] Ca-products-dad436e4c75d

red-optical · Kitt Peak (Kurucz 2005) · Ca I · reference lines · 1D LTE · [1], [2], [9], [3] Ca-products-3bba02358a4c

red-optical · Kitt Peak (Kurucz 2005) · Ca I · reference lines · 1D NLTE (Bergemann) · [1], [2], [9], [7] Ca-products-3e1e0b616e66

red-optical · Kitt Peak (Kurucz 2005) · Ca I · reference lines · 1D NLTE (Gerber) · [1], [2], [9], [3] Ca-products-bc4a6118b324

red-optical · Kitt Peak (Kurucz 2005) · Ca I · reference lines · 1D LTE (Gerber) · [1], [2], [9], [3] Ca-products-523bdf6b31f2

red-optical · Kitt Peak (Kurucz 2005) · Ca I · reference lines · <3D> LTE (Gerber) · [1], [2], [9], [3], [4] Ca-products-65cdf673c79b

red-optical · Kitt Peak (Kurucz 2005) · Ca I · reference lines · <3D> NLTE (Gerber) · [1], [2], [9], [3], [4] Ca-products-37a2e064a1ad

red-optical · Kitt Peak (1984, molecfit) · Ca I · Codex-graded lines · 1D LTE · [1], [2], [9], [3] Ca-products-32df2499bdd3

red-optical · Kitt Peak (1984, molecfit) · Ca I · Codex-graded lines · 1D NLTE (Bergemann) · [1], [2], [9], [7] Ca-products-2c5eab6e3fa0

red-optical · Kitt Peak (1984, molecfit) · Ca I · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [9], [3] Ca-products-33927267399d

red-optical · Kitt Peak (1984, molecfit) · Ca I · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [9], [3] Ca-products-19f451c52e7b

red-optical · Kitt Peak (1984, molecfit) · Ca I · Codex-graded lines · <3D> LTE (Gerber) · [1], [2], [9], [3], [4] Ca-products-0d3cb3c95ef2

red-optical · Kitt Peak (1984, molecfit) · Ca I · Codex-graded lines · <3D> NLTE (Gerber) · [1], [2], [9], [3], [4] Ca-products-affe075da451

red-optical · Kitt Peak (1984, molecfit) · Ca I · reference lines · 1D LTE · [1], [2], [9], [3] Ca-products-2dc922781b65

red-optical · Kitt Peak (1984, molecfit) · Ca I · reference lines · 1D NLTE (Bergemann) · [1], [2], [9], [7] Ca-products-a2bf8e0875d8

red-optical · Kitt Peak (1984, molecfit) · Ca I · reference lines · 1D NLTE (Gerber) · [1], [2], [9], [3] Ca-products-9a0319227190

red-optical · Kitt Peak (1984, molecfit) · Ca I · reference lines · 1D LTE (Gerber) · [1], [2], [9], [3] Ca-products-ec6084609ab2

red-optical · Kitt Peak (1984, molecfit) · Ca I · reference lines · <3D> LTE (Gerber) · [1], [2], [9], [3], [4] Ca-products-2fd32f084baa

red-optical · Kitt Peak (1984, molecfit) · Ca I · reference lines · <3D> NLTE (Gerber) · [1], [2], [9], [3], [4] Ca-products-66d76a7ec2b8

near-UV · Kitt Peak (Kurucz 2005) · Ca I · reference lines · 1D LTE (Gerber) · [1], [2], [9], [3] Ca-quarantine-d782099edd6b

near-UV · Kitt Peak (Kurucz 2005) · Ca I · reference lines · <3D> NLTE (Gerber) · [1], [2], [9], [3], [4] Ca-quarantine-a34a55f6f94b

near-UV · Kitt Peak (Kurucz 2005) · Ca I · reference lines · <3D> LTE (Gerber) · [1], [2], [9], [3], [4] Ca-quarantine-437509526dea

near-UV · Kitt Peak (Kurucz 2005) · Ca I · reference lines · 1D NLTE (Gerber) · [1], [2], [9], [3] Ca-quarantine-2521ae1fcb73

near-UV · Kitt Peak (Kurucz 2005) · Ca I · reference lines · 1D LTE · [1], [2], [9], [3] Ca-quarantine-f57c87504b38

near-UV · Kitt Peak (1984, molecfit) · Ca I · reference lines · 1D LTE (Gerber) · [1], [2], [9], [3] Ca-quarantine-fd39c9ff0a1f

near-UV · Kitt Peak (1984, molecfit) · Ca I · reference lines · <3D> NLTE (Gerber) · [1], [2], [9], [3], [4] Ca-quarantine-e64ed563a819

near-UV · Kitt Peak (1984, molecfit) · Ca I · reference lines · <3D> LTE (Gerber) · [1], [2], [9], [3], [4] Ca-quarantine-029460f2b3bd

near-UV · Kitt Peak (1984, molecfit) · Ca I · reference lines · 1D NLTE (Gerber) · [1], [2], [9], [3] Ca-quarantine-1f4d882cbccb

near-UV · Kitt Peak (1984, molecfit) · Ca I · reference lines · 1D LTE · [1], [2], [9], [3] Ca-quarantine-99e20e954b77

VIS · Kitt Peak (Kurucz 2005) · Ca I · reference lines · 1D LTE (Gerber) · [1], [2], [9], [3] Ca-quarantine-6f388cf8cf97

VIS · Kitt Peak (Kurucz 2005) · Ca I · reference lines · <3D> NLTE (Gerber) · [1], [2], [9], [3], [4] Ca-quarantine-f0fee6915ada

VIS · Kitt Peak (Kurucz 2005) · Ca I · reference lines · <3D> LTE (Gerber) · [1], [2], [9], [3], [4] Ca-quarantine-ead2807b8c53

VIS · Kitt Peak (Kurucz 2005) · Ca I · reference lines · 1D NLTE (Gerber) · [1], [2], [9], [3] Ca-quarantine-a8e49ddb4856

VIS · Kitt Peak (Kurucz 2005) · Ca I · reference lines · 1D LTE · [1], [2], [9], [3] Ca-quarantine-e5a1a2f0bd03

VIS · Kitt Peak (Kurucz 2005) · Ca I · reference lines · 3D NLTE (Amarsi) · [1], [2], [9] Ca-quarantine-fe262a330624

VIS · Kitt Peak (Kurucz 2005) · Ca I · reference lines · 1D NLTE (Bergemann) · [1], [2], [9] Ca-quarantine-792f4e8e8fc8

VIS · Kitt Peak (1984, molecfit) · Ca I · reference lines · 1D LTE (Gerber) · [1], [2], [9], [3] Ca-quarantine-54413fdaf191

VIS · Kitt Peak (1984, molecfit) · Ca I · reference lines · <3D> NLTE (Gerber) · [1], [2], [9], [3], [4] Ca-quarantine-6dec63fed77c

VIS · Kitt Peak (1984, molecfit) · Ca I · reference lines · <3D> LTE (Gerber) · [1], [2], [9], [3], [4] Ca-quarantine-8c904b9a34c7

VIS · Kitt Peak (1984, molecfit) · Ca I · reference lines · 1D NLTE (Gerber) · [1], [2], [9], [3] Ca-quarantine-3dead4c3e748

VIS · Kitt Peak (1984, molecfit) · Ca I · reference lines · 1D LTE · [1], [2], [9], [3] Ca-quarantine-07f83405f300

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